

Nonlinear Radiative Williamson Fluid Against a Wedge with Aligned Magnetic Field



K. Subbarayudu, L. Wahidunnisa, S. Suneetha, and P. Bala Anki Reddy

Abstract The foremost importance of this presentation is to explore the nonlinear thermal radiation on a Williamson liquid model on a wedge in the company of a heat generation/absorption which is not uniform. An aligned magnetic field, Brownian diffusion and thermophoresis aspects are also investigated. The flow and temperature equations are derived and solved by Runge–Kutta based *MATLAB bvp4c solver*. Results for different flow characteristics are plotted through graphs and discussed in detail. The wall temperature raises as temperature ratio parameter increases and results in a deep penetration for temperature. The concentration of the species seems to be increased with Brownian diffusion and radiation.

Keywords Williamson fluid model · Wedge shape geometry · Aligned magnetic field · Nonlinear thermal radiation

1 Introduction

The contemporary era, researchers are doing many experimental and theoretical studies on the fluid flow and transformation of energy in the non-Newtonian fluid models that have significant applications in engineering, for instance, emulsions, lubricants, polymers, and nuclear fuel slurries. Some alive rheological models are Power law, Carreau, Jeffery, Williamson fluid, and so forth. Out of these, Williamson fluid model is a simple model to suggest the viscoelastic nature and shear thin out features which were introduced by Williamson [1] in 1929. The fluid flow and transfer of heat across wedge-shaped geometries are important in several engineering applications and also in fluid dynamics. Particularly such flows occur in aerodynamics, heat exchangers, geothermal industries, and so on. A number of surveys have been found considering Williamson wedge flow in Ref. [2–4]. The study of fluid past a wedge with MHD has vital applications in nuclear reactor cooling, MHD power generators and

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so forth. Several authors [5–8] examined the MHD effects over a wedge in various types of fluids in different situations. Electromagnetic radiation is the major reason for radiation heat transfer. While framing a system in the industry with a negligible temperature variation inside a fluid, a troublesome is observed. To avoid this complexity, researchers included an extra parameter nonlinear thermal radiation in place of linear thermal radiation. Brief information on this area is quoted in Refs. [9–12]. At present, we consider heat source/sink which is not uniform, i.e., dependent on space and temperature. Some related studies were carried out by some investigators [13–16].

It is noted by the literature survey that a little information is existing on the MHD Williamson fluid model flow via a wedge with heat source/sink which is not uniform and radiation which is nonlinear. Inspired by these uses, the present study examines the influence of the thermophoresis effect and Brownian motion of Williamson fluid model for a radiating flow over a wedge.

2 Description of Physical Model

A two-dimensional, viscous, non-Newtonian liquid with Williamson model over a wedge with time-dependent aligned magnetic field is considered. A non-uniform heat generation/absorption and thermal radiation which is not linear is added. $u_w(x, t')$ represents wedge stretching velocity and is referred by $u_w(x, t') = bx^m(1 - ct')^{-1}$ where b -stretching rate and c -stable value. The flow velocity at free stream, $u_e(x, t') = ax^m(1 - ct')^{-1}$, while, a and c are stable values more than zero with $0 \leq m \leq 1$, along the wedge axis. $\Omega = \beta\pi$ supposed to be the angle of the wedge and the pressure gradient β is given as $\beta = \frac{2m}{m+1}$. The x -axis is incident in the elongating direction and y -axis makes 90° angle outside from it. A geometrical configuration and schematic model of the present objective model is put on viewed in Fig. 1. To study the heat flow on the surface of wedge, the temperature, $T_w(x, t') =$

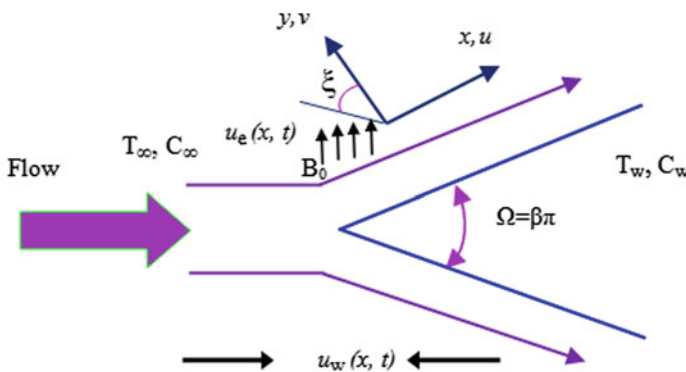


Fig. 1 Flow geometry

$\frac{T_0 u_w x}{v} (1 - ct')^{-\frac{1}{2}} + T_\infty$, concentration $C_w(x, t') = \frac{C_0 u_w x}{v} (1 - ct')^{-\frac{1}{2}} + C_\infty$, magnetic field $B(t') = B_0 (1 - ct')^{-\frac{1}{2}}$, T_0, C_0 be the initial temperature and concentration are considered and also, as y tends to ∞ , the free stream gain the constants values T_∞, C_∞ . With the above assumption, the equations are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$\begin{aligned} \frac{\partial u}{\partial t'} + \frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v = & \frac{\partial u_e}{\partial t'} + u_e \frac{\partial u_e}{\partial x} + v \frac{\partial^2 u}{\partial y^2} \left[\beta^* + (1 - \beta^*) \left(1 - \Gamma \frac{\partial u}{\partial y} \right)^{-1} \right] \\ & + v \Gamma \left(\frac{\partial u}{\partial y} \right) \frac{\partial^2 u}{\partial y^2} \left[(1 - \beta^*) \left(1 - \Gamma \frac{\partial u}{\partial y} \right)^{-2} \right] \\ & - \frac{\sigma B_0^2(t')}{\rho} \sin^2 \xi (u - u_e) \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial T}{\partial t'} + \frac{\partial T}{\partial x} u + \frac{\partial T}{\partial y} v = & \alpha \frac{\partial^2 T}{\partial y^2} - \frac{\partial q_r}{\partial y} \frac{1}{(\rho c_p)} \\ & + \tau \left[\frac{\partial T}{\partial y} \frac{\partial C}{\partial y} D_B + \frac{D_T}{T_\infty} \left(\frac{\partial T}{\partial y} \right)^2 \right] + \frac{q'''}{\rho c_p} \end{aligned} \quad (3)$$

$$\frac{\partial C}{\partial t'} + \frac{\partial C}{\partial x} u + \frac{\partial C}{\partial y} v = \frac{\partial^2 C}{\partial y^2} D_B + \frac{\partial^2 T}{\partial y^2} \frac{D_T}{T_\infty} \quad (4)$$

From Eq. (2) the fluid flow drive to viscous fluid case when $\beta^* = 0 = \Gamma$. The heat generation or absorption which is not uniform, is denoted as $q''' = [f'(T_w - T_\infty)A^* + (T - T_\infty)B^*] \frac{k u_w}{x v}$ where A^* —dependent relative to space and B^* —dependent relative to temperature of heat generation/absorption. The internal heat generation and absorption is noticed when $A^* > 0, B^* > 0$ and $A^* < 0, B^* < 0$.

For optically thick fluid, we use Rosseland approximation in which q_r denote radiative heat flux, is specified as

$$q_r = -(1.333) \frac{\sigma^*}{k^*} \frac{\partial T^4}{\partial y}, \quad (5)$$

We can set the term T^4 stirring in Eq. (5) and availing Taylor series by supposing an undersized deviation in temperature of the fluid and ambient fluid and within the boundary layer, preserving to first-order terms only. Hence T^4 referred to as

$$\frac{1}{4} T^4 \cong T_\infty^3 T - \frac{3}{4} T_\infty^4 \quad (6)$$

Equation (3) appears as Eq. (7) by utilizing Eqs. (5) and (6).

$$\begin{aligned} \frac{\partial T}{\partial t'} + \frac{\partial T}{\partial x} u + \frac{\partial T}{\partial y} v = \frac{\partial^2 T}{\partial y^2} \alpha + \frac{16\sigma^* T^3}{3(\rho c_p) k_1^*} \frac{\partial T}{\partial y} \\ + \tau \left[\frac{\partial C}{\partial y} \frac{\partial T}{\partial y} D_B + \left(\frac{\partial T}{\partial y} \right)^2 \frac{D_T}{T_\infty} \right] + \frac{q'''}{\rho c_p} \end{aligned} \quad (7)$$

related to the boundary conditions.

The wedge is assumed to be as there is no-slip condition on its surface and it follows:

On the shell of the wedge i.e., at $y = 0$

$$u = u_w, T = T_w, v = 0, C = C_w \quad (8)$$

At open water course i.e., as $y \rightarrow \infty$

$$u \rightarrow u_e, T \rightarrow T_\infty, C \rightarrow C_\infty \quad (9)$$

Using the quantities of non-dimensional [17]:

$$\begin{aligned} \eta = y \left(\frac{(m+1)u_e}{2vx} \right)^{\frac{1}{2}}, \psi = f(\eta) \left(\frac{2vxu_e}{m+1} \right)^{\frac{1}{2}}, \\ \phi(\eta) = \frac{C - C_\infty}{C_w - C_\infty}, \theta(\eta) = \frac{T - T_\infty}{T_w - T_\infty} \end{aligned} \quad (10)$$

at this juncture, the stream function $\psi(x, y, t)$ satisfies Eq. (1). The velocities are $u = \frac{\partial \psi}{\partial y}$ and $v = -\frac{\partial \psi}{\partial x}$. By using Eq. (10) in Eqs. (2), (7) and (4) we get the corresponding OD equations:

$$\begin{aligned} f''' \left[\beta^* + (1 - we f'')^{-2} (1 - \beta^*) \right] + f f'' + \left(\beta - \beta (f')^2 \right) \\ - (f' + \eta(0.5)f'' - 1)(2 - \beta)A - \text{Ha}^2 \{ f' - 1 \} (2 - \beta) \sin^2 \xi = 0, \end{aligned} \quad (11)$$

$$\begin{aligned} \frac{1}{\text{Pr}} \left[(1.333)\text{Rd} [1 + (\theta_w \theta - \theta)]^3 + 1 \right] \theta'' \\ + (1.333) \frac{\text{Rd}}{\text{Pr}} (\theta_w - 1) [(\theta_w \theta - \theta) + 1]^2 \theta'^2 + (f \theta' - 2 f' \theta) \\ - \frac{A}{2} (2 - \beta) (\eta \theta' + 3 \theta) + Nt \theta'^2 + Nb \phi' \theta' + (A^* f' + B^* \theta) = 0 \end{aligned} \quad (12)$$

$$\frac{1}{\text{Le}} \left[\phi'' + \frac{Nt}{Nb} \theta'' \right] - \frac{A}{2} (2 - \beta) (\eta \phi' + 3 \phi) - 2 \phi f' + f \phi' = 0, \quad (13)$$

with reduced conditions at boundary:

$$f = 0, f' = \lambda, \theta = 1, \phi = 1 \text{ at } \eta = 0 \quad (14)$$

$$f' \rightarrow 1, \theta \rightarrow 0, \phi \rightarrow 0 \text{ as } \eta \rightarrow \infty \quad (15)$$

Here, the wedge moving parameter, $\lambda = \frac{b}{a}$ and $\lambda > 0$: a stretching wedge, $\lambda < 0$: a shrinking wedge and $\lambda = 0$: a steady wedge.

The other engaged physical non-dimensional factors are:

$\beta^* = \frac{\mu_0}{\mu_\infty}$ is the ratio of viscosities, $\alpha = \frac{\kappa}{\rho c_p}$ the effective thermal diffusivity, $We = \sqrt{\frac{\Gamma^2(m+1)u_e^3}{2\nu x}}$ the Weissenberg number, $Pr = \frac{\mu c_p}{k}$ the Prandtl number, $Rd = \frac{4\sigma^* T_\infty^3}{kk_1^*}$ the Radiation parameter, $A = \frac{c}{ax^{m-1}}$ the unsteadiness parameter, $\beta = \frac{2m}{m+1}$ the wedge angle parameter, $\theta_w = \frac{T_w}{T_\infty}$ (>1) the temperature ratio parameter, $Ha^2 = \frac{\rho B_0^2}{\rho ax^{m-1}}$ the Hartmann number, $Le = \frac{\nu}{D_B}$ the Lewis number, $Nt = \frac{\tau D_T (T_w - T_\infty)}{T_\infty \nu}$ the thermophoresis parameter, $Nb = \frac{\tau D_B (C_w - C_\infty)}{\nu}$ the Brownian motion parameter.

The parameters which are useful in engineering are the local non-dimensional friction factor C_{f_x} , Nusselt number Nu_x and Sherwood number Sh_x . They are:

$$C_{f_x} = \frac{\tau_w}{\rho U_w^2}, Nu_x = \frac{x q_w}{k(T_w - T_\infty)} \text{ and } Sh_x = \frac{x q_m}{D_B(C_w - C_\infty)}$$

where τ_w , q_w and q_m are given as

$$\begin{aligned} \text{At } y = 0, \tau_w &= \mu_0 \left(\frac{\partial u}{\partial y} \right) \left[\beta^* + (1 - \beta^*) \left(1 - \Gamma \frac{\partial u}{\partial y} \right)^{-1} \right], \\ q_w &= [q_r] - \left[\frac{\partial T}{\partial y} \left(k + \frac{16\sigma T^3}{3k^*} \right) \right] \text{ and} \\ q_m &= -D_B \left(\frac{\partial C}{\partial y} \right) \end{aligned} \quad (16)$$

Using (10) and (16), the dimensionless Engineering quantities are given by

$$\begin{aligned} C_{f_x} Re_x^{\frac{1}{2}} &= \frac{1}{\sqrt{2-\beta}} \left[\beta^* + \{1 - we f''(0)\}^{-1} (1 - \beta^*) \right] f''(0), \\ Nu_x Re_x^{\frac{1}{2}} &= -\frac{1}{\sqrt{2-\beta}} \left(1 + \frac{4}{3} Rd [(\theta_w - 1)\theta(0) + 1]^3 \right) \theta'(0) \text{ and} \\ Sh_x Re_x^{\frac{1}{2}} &= -\frac{1}{\sqrt{2-\beta}} \phi'(0), \end{aligned} \quad (17)$$

where Reynolds number $Re_x = \frac{u_e x}{\nu}$.

3 Results and Discussions

The set of Eqs. (11)–(13) with (14) and (15) have been solved using the MATLAB bvp4c solver. This section analyzes the effects of parameters $A = 1.0$, $We = 0.5$, $\beta = 0.1$, $\beta^* = 0.2$, $Nt = 0.5$, $\theta_w = 1.1$, $Nb = 0.5$, $Le = 0.5$, $\lambda = 0.1$, $Rd = 0.5$, $Ha = 0.5$, $A^* = -0.05$, $B^* = -0.05$, $\xi = 45^\circ$, $Pr = 7.2$ associated with the flow problem. All the above values are kept steady unless mentioned in the figure. The mathematical results are plotted in diagrams and tables.

To attest to the genuineness of the attained results, an assessment is made with Hamid et al. [3] (Table 1). A wonderful agreement with the results is noticed. The skin friction values for diverse factors are submitted in Table 2. The development in the enormity of the skin friction is noticed as A , β , Ha and ξ increases. A reverse trend is noticed for λ . Table 3 provides the impact of numerous physical factors on Nusselt number. A rise in Nusselt number is observed as Pr , A rises. A fall in Nusselt number is noticed for a rise in Nb , Nt and We . Table 4 put on view the inspiration

Table 1 Comparison of wall friction coefficient for assorted β whilst $A = Ha = \beta^* = We = \lambda = 0$

β	Hamid et al. [3]	Present study
0.0	0.469600	0.4696
0.1	0.587035	0.5869
0.3	0.774755	0.7747
0.5	0.927680	0.8543
0.9	1.232588	0.9392

Table 2 Variations in the skin friction coefficient for variant values of A , β , λ , Ha and ξ when $We = 0.5$, $\beta^* = 0.2$, $\theta_w = 1.1$, $Rd = 0.5$, $Nt = 0.5$, $Nb = 0.5$, $Le = 0.5$, $A^* = -0.05$, $B^* = -0.05$, $Pr = 7.2$

A	β	λ	Ha	ξ	$-f''(0)$
1.0	0.1	0.1	0.5	$\pi/4$	0.3730
2.0	0.1	0.1	0.5	$\pi/4$	0.5370
3.0	0.1	0.1	0.5	$\pi/4$	0.6461
1.0	0.2	0.1	0.5	$\pi/4$	0.3804
1.0	0.3	0.1	0.5	$\pi/4$	0.3877
1.0	0.1	0.2	0.5	$\pi/4$	0.3114
1.0	0.1	0.3	0.5	$\pi/4$	0.2434
1.0	0.1	0.1	1.0	$\pi/4$	0.4638
1.0	0.1	0.1	1.5	$\pi/4$	0.5758
1.0	0.1	0.1	0.5	$\pi/4$	0.3899
1.0	0.1	0.1	0.5	$\pi/4$	0.4060

Table 3 Variations in the Nusselt number for variant values of Pr , Rd , θ_w , A^* , B^* , A , Nb , Nt , and We when $\beta = 0.1$, $\beta^* = 0.2$, $Le = 0.5$, $\lambda = 0.1$, $Ha = 0.5$, $\xi = 45^\circ$

Pr	Rd	θ_w	A	Nb	Nt	We	$\theta'(0)$
7.2	0.5	1.1	1.0	0.5	0.5	0.5	2.6121
10.0	0.5	1.1	1.0	0.5	0.5	0.5	2.8462
15.0	0.5	1.1	1.0	0.5	0.5	0.5	3.0025
7.2	1.0	1.1	1.0	0.5	0.5	0.5	4.0137
7.2	1.5	1.1	1.0	0.5	0.5	0.5	5.5615
7.2	0.5	1.2	1.0	0.5	0.5	0.5	3.1535
7.2	0.5	1.3	1.0	0.5	0.5	0.5	3.8511
7.2	0.5	1.1	2.0	0.5	0.5	0.5	3.7261
7.2	0.5	1.1	3.0	0.5	0.5	0.5	4.6236
7.2	0.5	1.1	1.0	1.0	0.5	0.5	2.2656
7.2	0.5	1.1	1.0	2.0	0.5	0.5	1.9774
7.2	0.5	1.1	1.0	0.5	1.0	0.5	2.2710
7.2	0.5	1.1	1.0	0.5	2.0	0.5	1.9390
7.2	0.5	1.1	1.0	0.5	0.5	1.0	2.6036
7.2	0.5	1.1	1.0	0.5	0.5	2.0	2.6003

Table 4 Variations in the Sherwood number for variant values of β , Le , A , Nb , Nt and We when $\beta^* = 0.2$, $\theta_w = 1.1$, $Rd = 0.5$, $Le = 0.5$, $\lambda = 0.1$, $Ha = 0.5$, $A^* = -0.05$, $B^* = -0.05$, $\xi = 45^\circ$, $Pr = 7.2$

β	Le	A	Nb	Nt	We	$\phi'(0)$
0.1	0.5	1.0	0.5	0.5	0.5	0.7887
0.2	0.5	1.0	0.5	0.5	0.5	0.8450
0.3	0.5	1.0	0.5	0.5	0.5	0.9038
0.1	0.6	1.0	0.5	0.5	0.5	0.8372
0.1	0.7	1.0	0.5	0.5	0.5	0.8815
0.1	0.5	2.0	0.5	0.5	0.5	0.2527
0.1	0.5	3.0	0.5	0.5	0.5	0.0412
0.1	0.5	1.0	1.0	0.5	0.5	1.2178
0.1	0.5	1.0	1.5	0.5	0.5	1.3409
0.1	0.5	1.0	0.5	1.0	0.5	1.0334
0.1	0.5	1.0	0.5	1.5	0.5	2.0021
0.1	0.5	1.0	0.5	0.5	1.0	0.6514
0.1	0.5	1.0	0.5	0.5	1.5	0.5401

of different physical factors on Sherwood number. A hike in Sherwood number is observed as β , Le , Nt , and Nb rise and an opposite trend is noticed for A and We .

The inspiration of Ha on velocity is shown in Fig. 2. In point of physics, hydro-magnetic flow for positive values of Ha and neutral for hydrodynamic flow. As of Fig. 2, it is noted an enhancement in Ha enhances the velocity. Influence of A on velocity is plotted in Fig. 3. With higher unsteadiness parameter, the fluid velocity

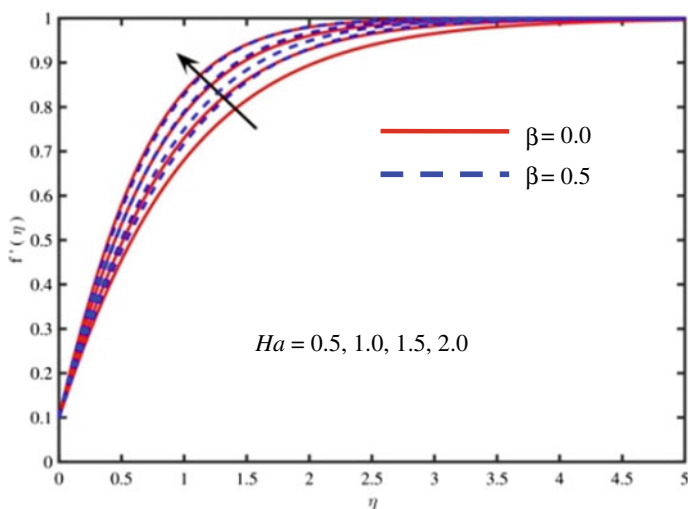


Fig. 2 $f'(\eta)$ via Ha

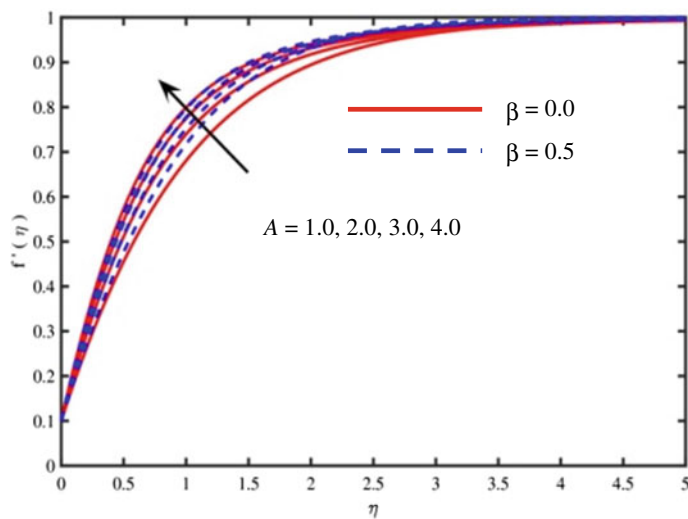


Fig. 3 $f'(\eta)$ via A

demonstrates an escalate behaviour near the boundary whereas a reverse trend is noticed for temperature. Figure 4 displays the upshot of Pr on temperature distribution over a wedge. Fluid temperature simply reduces for elevated Pr . As Pr promoted the rate of thermal diffusion is demoted. That is, accelerating Pr leads a loss in the boundary layer width of temperature. It is also noted that the thermal energy is high for $\beta = 0.5$ case compared with the other case. The outcome of thermophoresis on temperature is captured in 5. From Fig. 5, it is revealed that the temperature in the

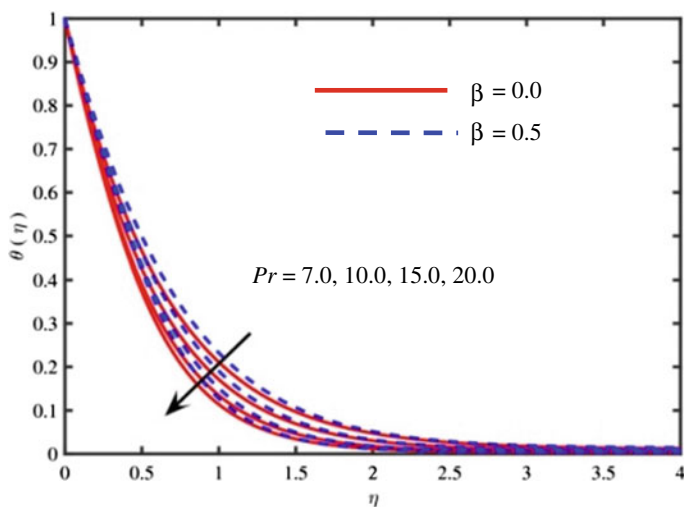


Fig. 4 $\theta(\eta)$ via Pr

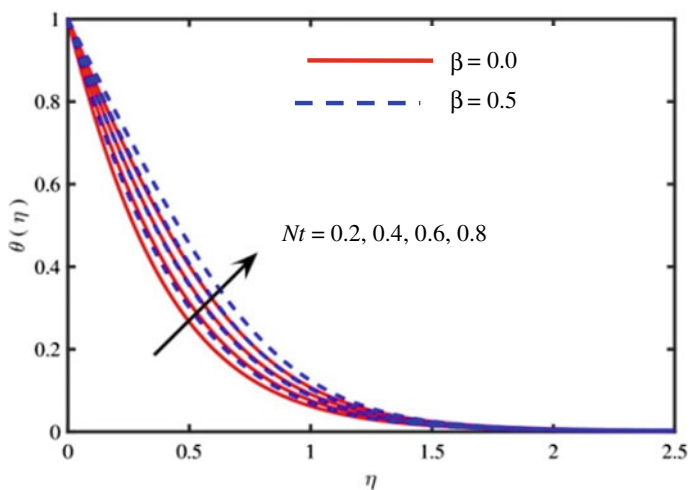


Fig. 5 $\theta(\eta)$ via Nt

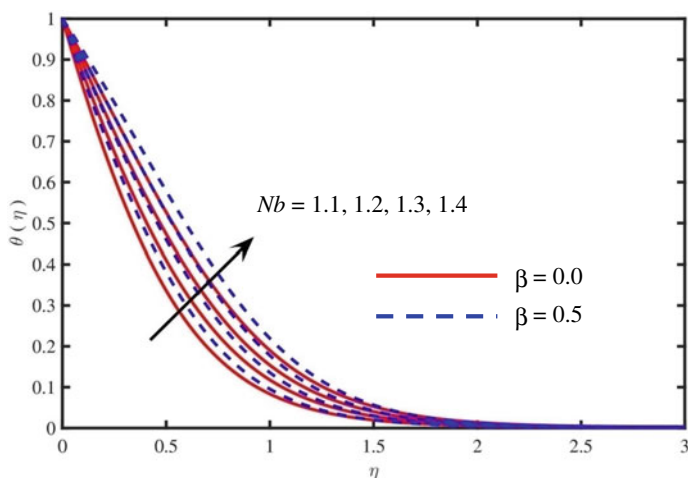


Fig. 6 $\theta(\eta)$ via Nb

boundary layer has increasing behaviour with an enhancement of Nt . In physics, it is a force generated by a tiny particle applying a force physically on another particle so that it moves from hot phase to a cold phase which frutage a thicker boundary layer for thermal and species. Therefore, far above Nt match up to elevated thermophoretic force because of temperature gradient, which promotes a speedy flow far from the wedge. Figure 6 divulge temperature distribution for various values of Nb . It can be viewed from Fig. 6 that the temperature is the proportion with Nb . The reason is that the Nb exhibits heat conduction. In addition, an enhancement in the Brownian motion heightens the random movement of the particles thus the width of the border layer grows for temperature. Figure 7 shows the impact of θ_w on temperature. The wall temperature raises as θ_w increases and results in a deep penetration for temperature. At high temperatures, the border layer for a temperature becomes bulky in the vicinity of the wall and skeletal far away from the sheet for low temperatures results in a modulation point at the surface of the wall for larger θ_w . Figure 8a, b depicts Nusselt number and Sherwood number for θ_w . Enhancement of θ_w enhances both the numbers for $\beta = 0.0$, and $\beta = 0.5$. Rising of the rate of heat transfer is observed from Figs. 9 and 10 which results in a growth in A^* and B^* for $\beta = 0.0$, and $\beta = 0.5$.

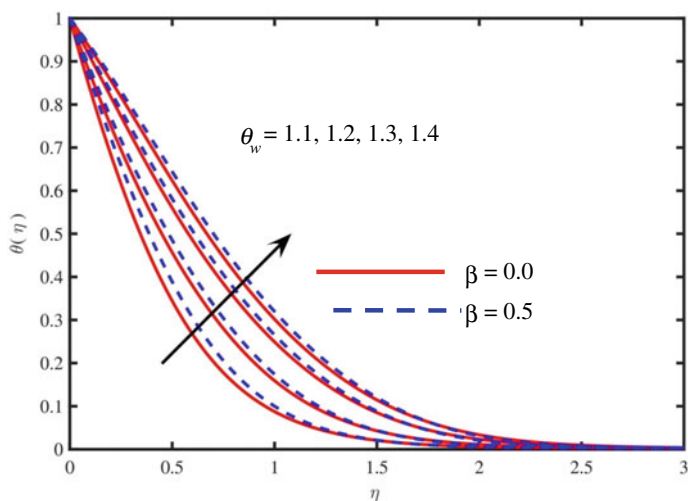


Fig. 7 $\theta(\eta)$ via θ_w .

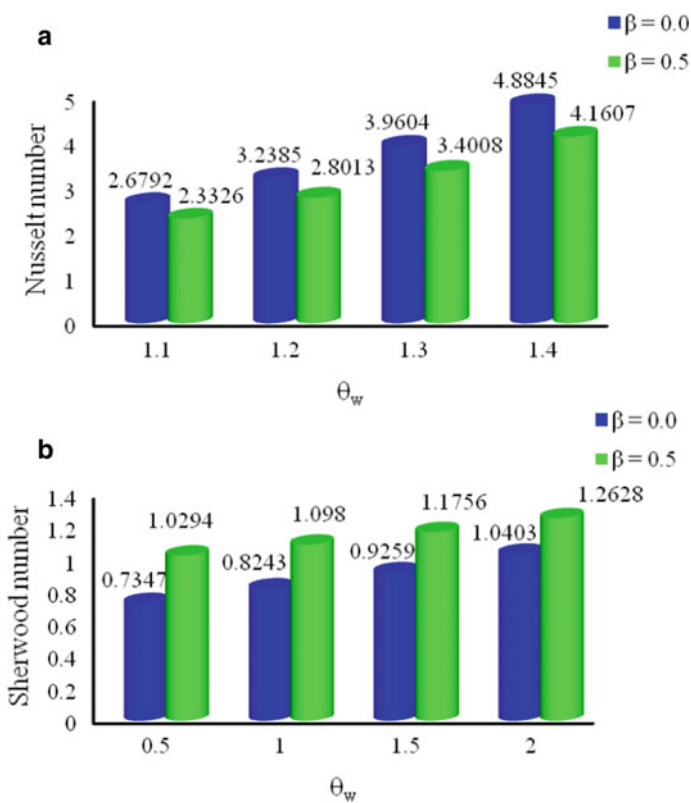


Fig. 8 a Nusselt number via θ_w , b Sherwood number via θ_w

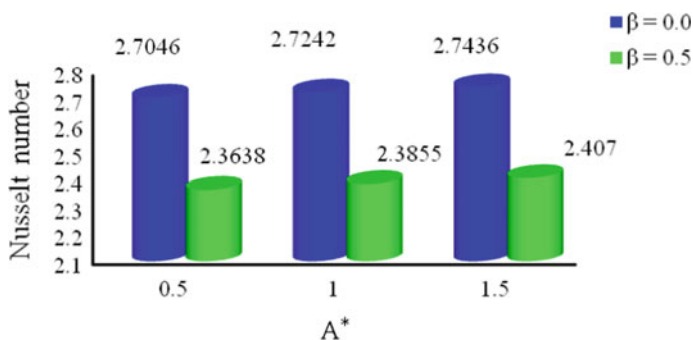


Fig. 9 Nusselt number via A^*

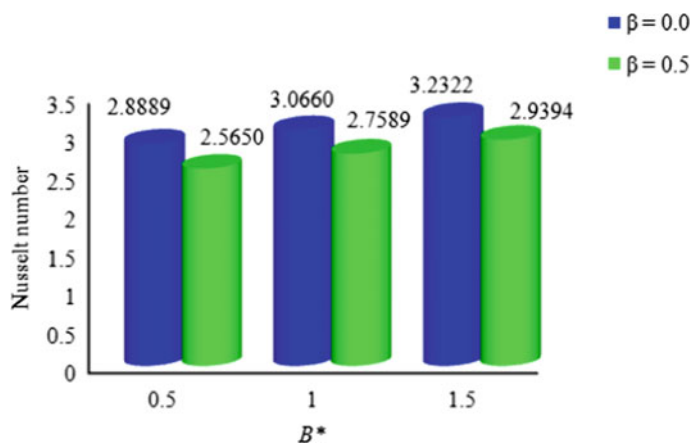


Fig. 10 Nusselt number via B^*

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